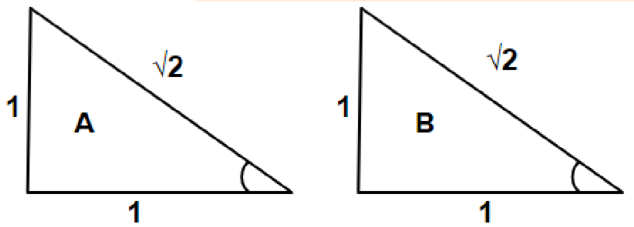
Telangana – Grade 10
2017 – Mathematics

Maximum Marks:40

1. If $\sin A = \frac{1}{\sqrt{2}}$ and $\cot B = 1$, prove that $\sin(A+B) = 1$, where A and B both are acute angles.

Solution:

$\sin(A+B) = \sin A \cos B + \cos A \sin B$



So, from figure

$$\sin(A+B) = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$$
$$= \frac{1}{2} + \frac{1}{2} = 1$$

2. The length of the minute hand of a clock is 3.5 cm. Find the area swept by minute hand in 30 minutes. (use $\pi = \frac{22}{7}$)

Solution:

Angle made in 30 minutes = $\frac{360}{60} \times 30 = 180°$

Area swept $= \frac{\theta}{360} \times \pi r^2$

$$= \frac{180}{360} \times \frac{22}{7} \times 3.5 \times 3.5$$
$$= 19.25 cm^2$$

3. Express $\cos\theta$ in terms of $\tan\theta$.

Solution:

$$\sec^2\theta - \tan^2\theta = 1$$
$$\sec^2\theta = 1 + \tan^2\theta$$
$$\frac{1}{\cos^2\theta} = 1 + \tan^2\theta$$
$$\cos^2\theta = \frac{1}{1+\tan^2\theta}$$
$$\cos\theta = \sqrt{\frac{1}{1+\tan^2\theta}}$$

4. From the first 50 natural numbers, find the probability of randomly selected numbers is a multiple of 3.

Solution:

$\text{Event} = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48\}$

$$n(s) = 50$$
$$p = \frac{16}{50} = \frac{8}{25}$$

5. Write the formula to find curved surface area of a cone and explain each term in it.

Solution:

C.S.A of cone = πrl

r – radius

l – slant height, $l = \sqrt{h^2 + r^2}$

6. "The median of observations, -2, 5, 3, -1, 4, 6 is 3.5". Is it correct? Justify your answer.

Solution:

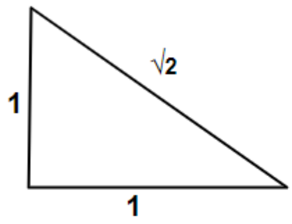
-1, -2, 3, 4, 5, 6

Median = $\frac{3+4}{2} = 3.5$

Hence, yes median = 3.5

7. If $\cos\theta = \frac{1}{\sqrt{2}}$, then find the value of $4 + \cot\theta$.

Solution:



$4+\cot\theta$

$\cot\theta = 1(\,from\ fig())$

$4+\cot\theta$

$=4+1=5$

8. The diameter of a solid sphere is 6 cm. It is melted and recast into a solid cylinder of height 4 cm. Find the radius of cylinder.

Solution:

Volume of sphere = volume of cylinder

$\frac{4}{3}\pi r^3 = \pi r^2 h$

$\frac{4}{3}\times 3\times 3\times 3 = r^2\times 4$

$9 = r^2$

$r = 3$

9. Write the formula of mode for grouped data, and explain each term in it.

Solution:

Mode $= l+\left(\frac{f_1-f_0}{2f_1-f_0-f_2}\right)\times h$

l – lower limit of modal class

h – size of class interval

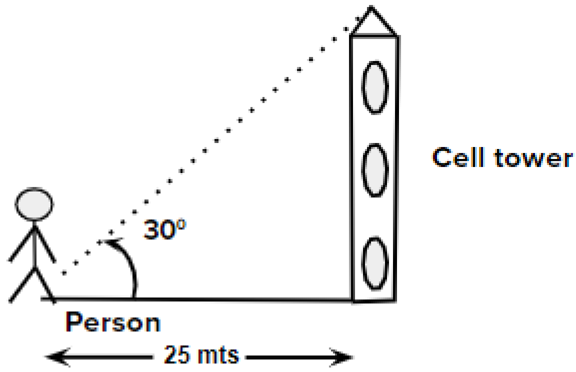
f_1 – frequency of modal class

f_0 – frequency of class preceding the modal class

f_2 – frequency of succeeding the modal class

10. A person 25 mts away from a cell - tower observes the top of cell - tower at an angle of elevation 30°. Draw the suitable diagram for this situation.

Solution:

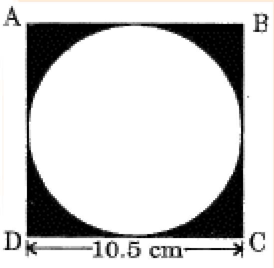


Cell tower = $\tan 30 \times 25$

$= \frac{1}{\sqrt{3}} \times 25$

$= \frac{25}{\sqrt{3}}$

11. Find the area of the shaded region in the given figure. ABCD is a square of side 10.5 cm.



Solution:

Area of square ABCD = 10.5×10.5

= 110.25 cm^2

Area of circle $= \frac{\pi d^2}{4}$

$= \frac{3.14 \times 10.5^2}{4} = 86.546 cm^2$

Area of shaded = 110.25 – 86.54

= 23.703 cm^2

12. One card is selected from a well- shuffled deck of 52 cards. Find the probability of getting a red card with prime number.

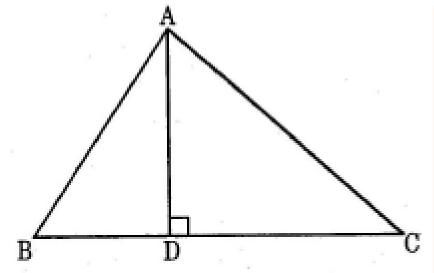
Solution:

Event $=\{(3,5,7)\,hearts+\{3,4,7\}diamonds\}$

$n(s)=52$

$p=\dfrac{6}{52}=\dfrac{3}{26}$

13. In a ΔABC, $AD\perp BC$ and $AD^2=BD\times CD$, prove that ΔABC is a right angled triangle.



Solution:

ΔABD is right angled

So, $AB^2=AD^2+BD^2$ (1)

ΔADC is also right angled

$AC^2=AD^2+CD^2$ (2)

Adding (1) and (2)

$$\begin{aligned} AB^2+AC^2 &= 2AD^2+BD^2+CD^2 \\ &= 2BD\times CD+BD^2+CD^2 \\ &= (BD+CD)^2 \end{aligned}$$

$AB^2+AC^2=BC^2$

Which is satisfied for ΔABC, to right angled at A.

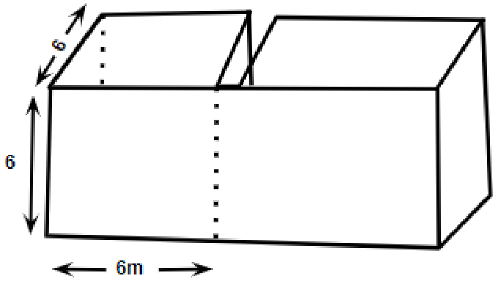
14. The length of cuboid is 12 cm, breadth and height are equal in measurements, and its volume is 432 cm^3. The cuboid is cut into 2 cubes. Find the lateral surface area of each cube.

OR

Two poles are standing opposite to each other on the either side of the road which is 90 feet wide. The angle of elevation from bottom of first pole to top of second pole is 45°, the angle of elevation from bottom of second pole to top of first pole is 30°. Find the heights of poles. (use $\sqrt{3}$ = 1.732)

Solution:

(a)



Let height = breath = x

Volume = 432

lbh = 432

$12 \times x \times x = 432$

$x^2 = \frac{432}{12}$

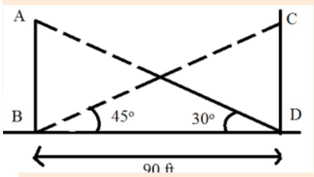
$x = 6cm$

$L.S.A of\ cube = 2(lb + bh + hl)$

$= 6s^2$

$= 6 \times 6^2 = 216cm^2$

(b)



In ΔABD

$\tan 30° = \frac{AB}{BD}$

$AB = \frac{1}{\sqrt{3}} \times 90 = \frac{90}{\sqrt{3}} f$

In ΔBCD

$\tan 45° = \frac{CD}{BD}$

$CD = 1 \times 90 = 90 f$

15. A bag contains some square cards. A prime number between 1 and 100 has been written on each card. Find the probability of getting a card that the sum of the digits of prime number written on it, is 8.

OR

The daily wages of 80 workers of a factory.

Daily Wages (Rs)	500-600	600-700	700-800	800-900	900-1000
Number of Workers	12	17	28	14	9

Find the mean wages of the works of the workers of the factory by using an appropriate method.

Solution:

$n(s)=\{2,3,5,7,11,13,17,19,23,29,31,37,39,41,43,47,53,59,61,67,71,73,79,83,89,97\}$

$n(sum)=\{(1,7)(5,3)(7,1)\}$

$p=\frac{3}{25}$

(b)

Daily wage	**Class marks** $\bar{x}$	**Number of workers f**	$\bar{\alpha} f$
500-600	550	12	6600
600-700	650	17	11050
700-800	750	28	21000
800-900	850	14	11900
900-1000	950	9	8550
		80	59100

Mean= $\frac{\sum a\bar{x}}{\sum_f}=\frac{59100}{80}=738.75$

16. Draw a circle of diameter 6 cm from a point 5 cm away from its centre. Construct the pair of tangents to the circle and measure their length.

OR

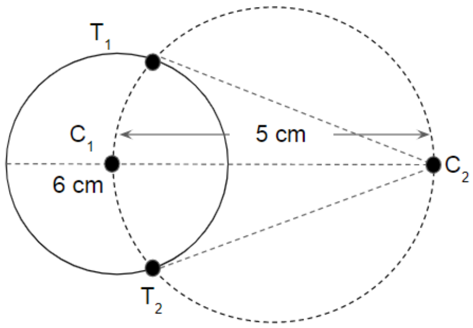
The following data gives the information on the observed life span (in hours) of 90 electrical components.

Life span (in hours)	0-20	20-40	40-60	60-80	80-100	100-120
Frequency	8	12	15	23	18	14

Draw both Ogives for the above data.

Solution:

(a)



Draw circle C_1 of 6 cm diameter.

Choose point P@ 5cm from center.

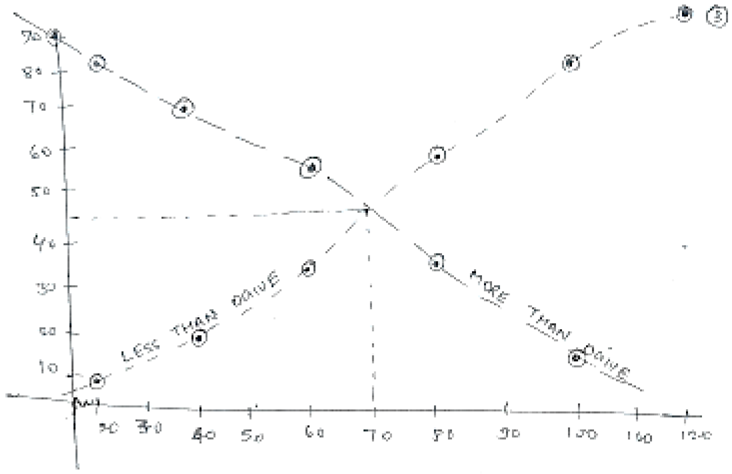
Draw circle C_2 with 5 cm diameter

Whenever C_2 cuts C_1 name as T_1 and T_2 measure its length.

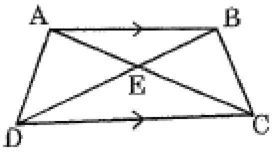
$T_1C_2=T_2C_2=4.00$ cm

(b)

Life span	**Frequency**	**Less than**	**Cum.f**	**More than**	**Cum.f**
0-20	8	Less than 20	8	More than 100	14
20-40	12	Less than 40	20	More than 80	32
40-60	15	Less than 60	35	More than 60	55
60-80	23	Less than 80	58	More than 40	70
80-100	18	Less than 100	76	More than 20	82
100-120	14	Less than 120	90	More than 0	90



17. ABCD is a trapezium with AB || DC, the diagonals AC and BD are intersecting at E. If ΔAED is similar to ΔBEC, then prove that $AD = BC$.



Prove that $(1+\tan^2\theta)+\left(1+\frac{1}{\tan^2\theta}\right)=\frac{1}{\sin^2\theta-\sin^4\theta}$.

Solution:

(a)

$\Delta AED \; and \; \Delta BEC$

$\angle AED = \angle BEC \; (vertically \; opposite)$

$In \; \Delta AEB \; and \; \Delta DEC$

$\angle AEB = \angle DEC \, (V.O.A)$

$\angle BAE = \angle ECD \, (Att.\text{int}\, terior \; angle)$

$\angle ABE = \angle CDE \, (Att.\text{int}\, erior \; angle)$

$so, \Delta AEB \cong \Delta DEC$

$hence \; AE = EC \; and \; DE = EB \; (C.P.Cf)$

Now,

$\Delta AED \; and \; \Delta BEC$

$\angle AED = \angle BEC \; (V.O.A)$

$AE = EC \; (proved)$

$DE = EB \, (prove.)$

So, $\Delta AEB \sim \Delta BEC$ by SAS property

Hence, AB = BC by corresponding side.

(b)

LHS

$$(1+\tan^2\theta)+\left(1+\frac{1}{\tan^2\theta}\right)$$

$$(1+\tan^2\theta)+\frac{\left(\tan^2\theta+1\right)}{\tan^2\theta}$$

$$=\frac{\tan^2\theta+\tan^4\theta+\tan^2\theta+1}{\tan^2\theta}$$

$$=\frac{\left(1+\tan^2\theta\right)^2}{\tan^2\theta}=\frac{\left(\sec^2\theta\right)^2}{\tan^2\theta}=\frac{1}{\cos^4\theta\times\sin^2\theta}=\frac{1}{\sin^2\theta\cos^2\theta}$$

$$=\frac{1}{\sin^2\theta(1-\sin^2\theta)}$$

$$=\frac{1}{\sin^2\theta-\sin^4\theta}=RHS$$